



ETHICAL TECH COLAB · NYU CENTER FOR GLOBAL AFFAIRS · RESEARCH REPORT

AFTER THE CORRIDOR

FROM AI-INFORMED EVACUATION TO DIGITAL PUBLIC GOODS FOR REFUGEE ECONOMIC INCLUSION

How can AI inform evacuation decisions, and what happens after? This report examines the CoLab's five fielded evacuation prototypes, their limitations and research frontiers, and extends the arc to the camp: the measurement, financial-modeling, and rights infrastructure that can shorten protracted displacement, grounded at Dzaleka Refugee Camp, Malawi.

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1. ABSTRACT & THE FUNDING CASE

The Ethical Tech CoLab's evacuation portfolio was built to answer a protection question: how can and should AI tools be used to enhance the protection and evacuation of civilians during armed conflict, and in what ways do they support or challenge existing obligations under International Humanitarian Law (IHL)?

Across five fielded prototypes — a crisis-information index, a unified risk platform, a calibrated cost-and-risk decision framework, an agent-based behavior model, and a readiness-under-uncertainty simulator — one finding recurred: the binding constraint on protection is rarely the absence of a route or a resource, but the quality, accessibility, and equity of the information people and responders act on. IHL frames evacuation as a last-resort duty¹; the technology's job is to make that duty implementable under uncertainty.

This report extends the arc past the corridor. A successful evacuation typically ends in a camp, where the modal outcome is years of protracted displacement. Using Dzaleka Refugee Camp, Malawi — where humanitarian financing has collapsed while an encampment policy bars residents from work — we show that the same measurement discipline, financial modeling, and uncertainty methods that served evacuation convert directly into durable-solutions infrastructure: an open Livelihoods Information & Access Index (LIAI), a resettlement-and-inclusion financial model built on the CoLab's operational evacuation cost engine, privacy-preserving cash and credential rails, and a civil-society legal-assistance map. We also state what we will not build — biometric identity for displaced people — and why that refusal is itself a public good.

THE ASK, IN ONE PARAGRAPH

The CoLab seeks grant funding to deploy this technology infrastructure as open public goods with displaced communities — validated first at Dzaleka with field partner Fraternidade Sem Fronteiras, and released as portable instruments any research or practitioner team can adopt. The proximate vehicle is the IPA/J-PAL Displaced Livelihoods Initiative (Round V, Infrastructure & Public Goods Creation); the architecture, team, measurement plan, and success criteria in this report are written to survive that diligence and any peer funder's.

2. THE RESEARCH QUESTION, AND WHY IT RUNS PAST THE CORRIDOR

The portfolio's organizing question comes from Teresa Cantero's doctoral research at the Instituto de Derechos Humanos "Gregorio Peces-Barba," Universidad Carlos III de Madrid: how can and should artificial intelligence tools be used to enhance the protection and evacuation of civilians during armed conflict, and in what ways do they support or challenge the implementation of existing obligations under IHL?² Under IHL, evacuation is not discretionary: when the security of the population or imperative military reasons so demand, moving civilians out of danger is a binding duty (GC IV art. 49; AP I arts. 57–58; AP II art. 17) — while a movement into which people are coerced is forcible transfer, a war crime regardless of motive. Any decision-support tool in this space sits on that line, and its design must keep evacuations protective rather than coercive.

Two structural facts motivate the program. First, a protection gap: states race to adopt AI for targeting and operational advantage, but rarely operationalize computational models to proactively

¹Geneva Convention Relative to the Protection of Civilian Persons in Time of War (Fourth Geneva Convention), art. 49; Protocol Additional to the Geneva Conventions (Protocol I), arts. 57–58; Protocol II, art. 17. See also "Evacuation." How Does Law Protect in War? ICRC Casebook, casebook.icrc.org/a_to_z/glossary/evacuation.

²Cantero, Teresa. "Artificial Intelligence and Civilian Protection in Evacuation during Armed Conflict under International Humanitarian Law." Presentation, NYU Center for Global Affairs, 2026.

protect civilians or make evacuation decisions legible under the incomplete, fast-changing information those decisions face.³ Second, an accounting gap: displacement is tracked exhaustively (117.3 million people displaced as of 2025), yet there is no global count of “evacuated civilians,” because evacuation is fragmented across militaries, governments, humanitarians, volunteers, and civilians themselves.⁴

The arc does not end when the shooting stops. An evacuation succeeds when a person reaches relative safety — usually a camp — and there the analytical problem inverts: from “should we move, and how,” to “now that we are here, how do we leave well.” The three durable solutions — voluntary return, local integration, resettlement — are each gated by information and legal status far more than by physical capacity. This report treats the durable-solutions phase as the natural continuation of the evacuation research question.

³Cantero, “Artificial Intelligence and Civilian Protection,” slides “The Protection Gap” and “Decision-Making Under Uncertainty.”

⁴UNHCR. Global Trends: Forced Displacement in 2024. UNHCR, 2025. Figures as cited in Cantero 2026.

3. THE EVACUATION PORTFOLIO: FIVE PRODUCTS, THEIR LIMITS, AND WHAT THEY OPEN

Each prototype is examined for method, honest limitations, and the research it opens. All are open source at github.com/Ethical-Tech-CoLab.

PROTOTYPE	FUNCTION	CORE METHOD
Evacuation Information Index (EII)	Scores quality, accessibility & equity of crisis information by geography	Composite index (INFORM/ACLED/FSI logic); live news + conflict feeds
Exodus — Civilian Evacuation Risk Platform	Unifies EII + cost model + endangerment/feasibility assessment	FastAPI backend; shared calculators & data layer
ERCF — Evacuation Risk & Cost Framework	Estimates human & financial cost of evacuating vs. staying	7-dimension scoring; calibrated cost model; break-even analysis
Evacuation Behavior Simulator	Models how information & demographics shape who leaves, and when	Agent-based model; probabilistic status lifecycle
Readiness & Uncertainty Simulator	Models how intelligence uncertainty degrades destination choices	Monte Carlo; gatekeeper factors; uncertainty slider

3.1 EVACUATION INFORMATION INDEX (EII)

The EII treats crisis information itself as the object of measurement, scoring Impact (20%), Conditions (50%), and Complexity (30%) on a 1–5 scale with weights synthesized from the INFORM Severity Index, ACLED, and the Fragile States Index; a live backend adds current news and an ACLED conflict timeline per crisis, and a map layer can overlay AI-derived building- and route-damage tiles from Microsoft HASTE + Planet imagery.⁵ The design is deliberately transparent: weights are labeled provisional pending Analytic Hierarchy Process (AHP) expert validation, and where the hosted site cannot run the live backend, responses are baked into a clearly labeled static snapshot — nothing is faked.

LIMITATIONS

- Weights are borrowed by analogy, not yet expert-validated or sensitivity-tested; a reviewer will read pre-validation weights as pre-evidence.
- Source and language bias: feeds skew to English-language, internationally indexed reporting — so the populations with the worst information access are least visible to the instrument that scores information access. The index must correct for this, not inherit it.
- The HASTE damage overlay is self-hosted with no public API; until connected, the flagship information- $\times$ -damage view is empty for most users.

FUTURE RESEARCH

Run the three-phase validation the CoLab's methodology work already specifies — Budget-Allocation + Delphi elicitation with 8–12 domain experts, Fuzzy-AHP on contested weights,

⁵Ethical Tech CoLab. "Evacuation Inform Index (EII)." GitHub, 2026, github.com/Ethical-Tech-CoLab/evacuation-inform-index-carolina. Weighting precedents: ACAPS / EU Joint Research Centre. INFORM Severity Index, acaps.org; ACLED. Conflict Index, acleddata.com; Fund for Peace. Fragile States Index, fragilestatesindex.org.

then historical case calibration⁶ — and add an equity layer measuring who receives information (gender, age, nationality, in/out-of-camp). That equity layer is the move that turns EII into the livelihoods instrument of §6.

3.2 EXODUS — CIVILIAN EVACUATION RISK PLATFORM

Exodus is the integration layer: one FastAPI application unifying the EII crisis map, the ERCF scenario and cost model, and a CERAI endangerment-plus-feasibility assessment behind a single backend, design language, and API surface — including calculators for staying-cost, evacuation-cost, resources, and scenario management.⁷ Integration is not validation: unifying three research prototypes inherits their uncalibrated assumptions behind a more authoritative interface, which raises the duty to label confidence. Its API is also the seam where the durable-solutions module of §5 plugs in.

3.3 ERCF — THE FINANCIAL MODELING ENGINE (OPERATIONAL)

ERCF v7.2 is the CoLab's calibrated decision-support tool for the human and financial cost of civilian evacuation in armed conflict — and it is the financial-modeling capability this proposal builds on, already in place and statistically validated in its evacuation form. It scores scenarios across seven dimensions (Kinetic Threat, Mobility, Authorization, Logistics, Destination, Urgency, Information), estimates evacuation resources and in-situ assistance costs, and produces a break-even analysis: the day after which a one-time evacuation becomes cheaper than the mounting daily cost of sustaining a population in place.⁸

Its cost parameters are sourced to humanitarian standards and individually tagged Validated, Estimated, or Unvalidated: water 20 L/person/day (UNHCR WASH Manual), food 0.45 kg/person/day and shelter/medical ratios (Sphere 2018), UNHAS air at US\$2.08/pax-km (WFP 2025), a WFP baseline of US\$0.42/person/day (WFP APR 2023).⁹ The mortality model is disclosed as indicative rather than predictive (R^2 log-log 0.855, LOOCV 0.807, with a documented structural failure case), while the financial estimates are substantially more reliable — which is why the durable-solutions adaptation of §5 leads with cost.

Ethical architecture. ERCF states explicitly that it does not place a monetary value on human life; its figures are logistics estimates, and its labels are descriptive, not imperative — it presents data to support a decision rather than prescribing one. This framing is carried verbatim into every extension in this report.

LIMITATIONS & FUTURE RESEARCH

- It models the immediate field operation only (~US\$134/person at moderate risk, 50 km, ground); real operations — Kosovo 1999 at ~US\$562/person, Lebanon 2006 at ~US\$4,500–5,700 — add international transport, reception, and hosting. Extending the cost boundary to the full displacement journey, including protracted encampment, is the frontier §5 takes up.

⁶Saaty, Thomas L. "How to Make a Decision: The Analytic Hierarchy Process." *European Journal of Operational Research*, vol. 48, no. 1, 1990, pp. 9–26; Al Fozaie, Mohammed. "A Guide to Integrating Expert Opinion and Fuzzy AHP When Generating Weights for Composite Indices." *Advances in Fuzzy Systems*, 2022, doi:10.1155/2022/3396862; OECD/JRC. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, 2008.

⁷Ethical Tech CoLab. "Exodus — Civilian Evacuation Risk Platform." GitHub, 2026, github.com/Ethical-Tech-CoLab/Exodus.

⁸Ethical Tech CoLab / Rocha, Yago. "ERCF — Evacuation Risk and Cost Framework, v7.2." GitHub, 2026, github.com/Ethical-Tech-CoLab/ercf.

⁹UNHCR. *WASH Manual*; Sphere Association. *The Sphere Handbook*. 2018 ed.; World Food Programme. *Annual Performance Report 2023*; WFP Executive Board figures 2025 — as compiled and tagged in the ERCF v7.2 cost-model documentation.

- Ground-truthing against OCHA's Financial Tracking Service is on the backlog and belongs in the funded work plan.

3.4 EVACUATION BEHAVIOR SIMULATOR

An agent-based model showing how information spread and demographics shape who leaves and when: agents move through an UNAWARE → SEEKING → MILLING → EVACUATING → DONE lifecycle, with family clusters that wait for all members, confirmation loops whose speed depends on information clarity, neighbor influence, and calibrated delays for elders, children under five, pregnant individuals, and unaccompanied minors.¹⁰ It is pedagogical rather than predictive — but its household-composition logic transfers directly to modeling who is left behind by information failures inside a camp.

3.5 READINESS & UNCERTAINTY SIMULATOR

A seeded, fully reproducible instrument that models how field-intelligence uncertainty degrades evacuation decisions: destinations are scored across seven factors, 500 Monte Carlo trials per group–destination pair convert readiness into a success-probability distribution, and an uncertainty slider degrades every factor's confidence at once — letting a user watch predicted outcomes collapse while nothing on the ground changes.¹¹

Two design choices carry the rest of this report. First, three factors are non-substitutable gatekeepers — Security, Local Authority Consent, and host-community Willingness — any of which hard-caps readiness: a destination that is dangerous, legally inaccessible, or whose hosts have refused cannot be redeemed by good logistics. Second, the model insists on the distinction between Unknown (a factor not yet assessed — a gap AI can close) and Unwilling (a confirmed refusal — a settled exclusion); conflating them would make an unresolvable political refusal look like a solvable data problem. Structurally, “assign groups to destinations, gated by consent, willingness, security, and capacity, under uncertainty” is resettlement matching — the transfer \$5 makes.

3.6 CROSS-CUTTING SYNTHESIS

1. Information access is the binding constraint — the thread that carries into the camp.
2. Model uncertainty explicitly, or mislead by omission: separate “unknown” from “confirmed,” always.
3. Composite indices are portable infrastructure: the INFORM → AHP → calibration pattern is domain-agnostic and reusable.
4. Descriptive, never prescriptive: ERCF's refusal to monetize life is the template for cost modeling in the refugee-lives context.
5. Artifact-first and reproducible: seeded, shareable, openly labeled outputs are what let this work face diligence.

¹⁰Ethical Tech CoLab / MacKew, Melanie. “Evacuation Simulation.” GitHub, 2026, github.com/Ethical-Tech-CoLab/Evac-Sim-Melanie.

¹¹Ethical Tech CoLab / Clarke, India. “Evacuation Readiness & Uncertainty Simulator.” GitHub, 2026, github.com/Ethical-Tech-CoLab/India-EvacSimulation.

4. AFTER ARRIVAL: DZALEKA AND THE SECOND EMERGENCY

A successful evacuation deposits people into a slower emergency. Most refugees are in protracted situations; return rates are low, resettlement reaches a small fraction, and care-and-maintenance encampment becomes the default by inertia. Dzaleka Refugee Camp in Dowa District, Malawi — built in 1994 for 10,000–12,000 people, now holding roughly 53,000–60,000, predominantly from the DRC, Burundi, and Rwanda — is the sharp end of the 2025–26 financing collapse:

- UNHCR's Malawi budget fell from about US\$8 million to about US\$1 million — a ~90% cut — ending general non-food distributions and laying off protection staff, including paralegals.¹²
- WFP assistance is secured only through mid-2026, with about US\$11 million still needed to restore full rations through December; monthly cash support has fallen toward US\$5.90–US\$8.60 per person, down from roughly US\$100, forcing skipped meals and negative coping.¹³
- Malawi's encampment policy, enforced since March 2023, bars most residents from working outside the camp — so the population cannot substitute self-reliance for collapsing aid. Local civil society (e.g. Inua Advocacy) argues the crisis is inseparable from that policy and is pressing for the right to work; weakened oversight is meanwhile reported to embolden trafficking networks.¹⁴

This is not a context for piloting speculative technology on vulnerable people. It is a context where the highest-value contribution is measurement, evidence, and legibility: tools that help residents and the organizations serving them see and argue for the fastest route out, and that document the true cost of the status quo. That is the design brief for §§5–7.

5. THE FINANCIAL MODELING TOOL: FROM EVACUATION BREAK-EVEN TO INCLUSION BREAK-EVEN

The financial modeling capability is already in place. ERCF (§3.3) is an operational, calibrated engine — exposed through Exodus's API as staying-cost and evacuation-cost calculators — whose core mathematics is exactly the comparison the durable-solutions question needs: a one-time investment versus a mounting recurring cost, resolved into a break-even day. The Resettlement & Inclusion Capacity Simulator (RICS) is that same engine re-pointed, not a new build:

- **Cost layer (ERCF, re-parameterized).** Trajectory A prices status-quo encampment as a recurring per-person, per-year assistance cost (rations, WASH, health, protection, administration — the same Sphere/UNHCR/WFP parameter families ERCF already carries, tagged Validated/Estimated/Unvalidated). Trajectory B prices an inclusion or resettlement pathway as a front-loaded investment (documentation, skills recognition, transport, start-up capital, host co-investment). Output: the break-even year after which the status quo costs more — ERCF's break-even chart with the axes relabeled.
- **Capacity & willingness layer (readiness simulator, transferred).** Destinations — resettlement states, local-integration pathways — are mapped against the non-substitutable

¹²"Dzaleka Refugees at Risk as WFP Faces K19bn Shortfall." Nation Online, 8 Feb. 2026, mwnation.com/dzaleka-refugees-at-risk-as-wfp-faces-k19bn-shortfall/; "Funding Cuts Push Malawi's Dzaleka Refugee Camp to the Brink." FairPlanet, 20 Aug. 2025, fairplanet.org.

¹³Nation Online, 8 Feb. 2026; "'We Still Need People to Hold Our Hands': Inside the Collapse of Malawi's Humanitarian Safety Net." Dialogue Earth, June 2026, dialogue.earth; UNHCR. "Refugees in Dzaleka Refugee Camp Struggle to Make Ends Meet amid Funding Cuts." UNHCR Africa, 5 May 2025.

¹⁴"Dzaleka Deadline Looms, Government's Response Plan Still Uncertain." The Daily Times (Malawi), June 2026, times.mw; FairPlanet, 20 Aug. 2025.

gatekeepers (legal Consent, host Willingness, Security) alongside substitutable capacity (housing, jobs, services), preserving Unknown-vs-Unwilling so an un-lobbied state is never mistaken for a closed door.

- **Uncertainty layer.** The uncertainty slider carries over, so thin data on host capacity or a state's posture visibly widens the confidence band — preventing false precision in advocacy numbers.

WHAT THE LITERATURE SAYS THE MODEL WILL FIND

The engine formalizes an argument the development-economics evidence already supports. UNHCR estimates that if freedom of movement and the right to work were relaxed so refugees could earn on par with hosts, complementary assistance costs would fall from about US\$3.2 billion to roughly US\$900 million a year; the joint World Bank–UNHCR benchmark puts refugees' global subsistence needs at US\$56–62 billion annually, of which about US\$41 billion is already met by refugees' own earnings where work is permitted.¹⁵ In Uganda's inclusive regime, economic participation saves an estimated US\$150 per refugee per year — about US\$225 million across 1.5 million refugees; around Kakuma, refugee presence raised gross regional product ~3.4%; Colombia's regularization of ~500,000 Venezuelans raised their incomes ~31% with minimal impact on host formal employment.¹⁶

The honest caveat is built in: the World Bank finds self-reliance “remains elusive” across much of Sub-Saharan Africa because camps often sit in marginal, high-poverty regions where even full local integration would lift incomes little.¹⁷ RICS must be able to return “inclusion does not pay here” where host markets are too thin — otherwise it is advocacy, not analysis. Its output is a defensible, reproducible sentence: sustaining this cohort at Dzaleka costs approximately X per year; this pathway costs approximately Y up front and breaks even in year Z, subject to these gatekeepers and this uncertainty band.

¹⁵UNHCR. “Investing in Refugees' Self-Reliance: A More Cost-Effective and Sustainable Response.” UNHCR Blog, unhcr.org/blogs/investing-in-refugees-self-reliance/.

¹⁶World Bank. “The Costs Come Before the Benefits.” World Bank Group, 2024, openknowledge.worldbank.org; “Refugees Mean Business: This Is Why Investing in Them Pays.” World Economic Forum, 20 Jan. 2025, weforum.org.

¹⁷World Bank. Making Refugee Self-Reliance Work: From Aid to Employment in Sub-Saharan Africa. World Bank, 2025.

6. DIGITAL PUBLIC GOODS FOR ECONOMIC INCLUSION

The toolkit, ordered from safest to most caution-demanding. The organizing principle is data minimization: build the thinnest system that solves the problem, and refuse the parts whose risks outweigh their benefit for this population (§7).

6.1 MEASUREMENT: THE LIVELIHOODS INFORMATION & ACCESS INDEX (LIAI)

LIAI is the EII re-specified for livelihoods and legal rights: an open, versioned instrument measuring the quality, accessibility, and equity of information about work rights, employment pathways, financial services, and entrepreneurship support reaching displaced populations — turning “information access,” usually an unmeasured assumption, into a decomposable, shareable variable that future impact evaluations can treat as outcome, mechanism, or moderator, disaggregated by gender, age, nationality, and in/out-of-camp status. Deliverables: the index, an open data standard, a reference implementation, and methodology documentation — validated first at Dzaleka with Fraternidade Sem Fronteiras.

6.2 VALUE TRANSFER: PRIVACY-PRESERVING CASH, DECOUPLED FROM BIOMETRICS

The nearest-term win is moving money more cheaply and reliably as traditional financing collapses. The reference case is WFP’s Building Blocks, the largest blockchain-based humanitarian platform: settling entitlements on a private, permissioned ledger and paying vendors in lump sums saved roughly 98% of transaction costs (about US\$2.4 million on an early deployment), improved cross-agency reconciliation, and established a co-governed cooperative network that UN Women extended to cash-for-skills in refugee camps.¹⁸

The same case is the cautionary tale: Building Blocks ran on biometric (iris) registration, and independent analysis warns that many such pilots functioned as donor “spectacles” that under-invested in privacy, where even a permissioned ledger leaks sensitive metadata if on-chain pseudonyms are correlated with off-chain records.¹⁹ The lesson is decouple the rail from biometric identity. Design rules for a Dzaleka wallet: authenticate with the thinnest credential (PIN plus locally issued token or QR), never a biometric; keep identifying data off-chain with only pseudonymous entitlement state on-ledger; integrate an existing audited, co-governed network rather than building a bespoke ledger — the CoLab’s value-add is the threat model and the measurement wrapper; and run a data-protection impact assessment with community consultation before any collection.

6.3 PORTABILITY: SKILLS CREDENTIALS — NEVER IDENTITY

The CoLab already operates the relevant architecture: its Arts Provenance Agent issues cryptographically signed, tamper-evident JSON-LD Verifiable Credentials (did:pkh, secp256k1) for objects²⁰, and Prof. Rhodes’s prior ID2020 work concerned decentralized, portable humanitarian credentials with UNHCR and WFP. The responsible human application is narrow: portable, self-held credentials for skills, qualifications, and completed training — so a nurse or electrician does not lose a decade of accreditation to displacement — issued to and controlled by the individual, presentable selectively. The bright line: a skills credential says what a person can do; it must never become a de

¹⁸World Food Programme. “Building Blocks.” WFP Innovation, innovation.wfp.org/project/building-blocks; UN Women and WFP. Blockchain Pilot Project for Cash Transfers in Refugee Camps: Jordan Case Study. UN Women, 2021; “How Blockchain Can Mobilize Emergency Resources for Humanitarian Assistance.” NYU SPS Global Affairs Review, 20 Mar. 2023.

¹⁹“Privacy in Peril: Safeguarding Digital Data in Humanitarian Blockchain Initiatives.” Humanitarian Practice Network / ODI, 27 Aug. 2025, odihpn.org.

²⁰Ethical Tech CoLab. “Arts Provenance Agent.” GitHub, 2026, github.com/Ethical-Tech-CoLab/arts-provenance-agent.

facto identity card that gatekeeps aid or is legible to a persecuting authority. The moment it holds who someone is, it inherits §7's risks and should be stopped.

6.4 RIGHTS: MAPPING CIVIL-SOCIETY LEGAL ASSISTANCE

Donor cuts stripped UNHCR's own paralegals out of Dzaleka — which makes a maintained, open directory of who actually provides legal help the safest, highest-leverage build. Per legal need (refugee-status determination, work authorization, documentation, GBV protection, anti-trafficking, resettlement referral), the map records which civil-society and refugee-led organizations provide assistance, with contact, language, and eligibility — under the same allowlist-and-citation discipline the provenance agent enforces (no unsourced claim ships). It plugs into LIAI: the index shows where rights information fails; the map routes a person to the organization that can close the gap. Local actors such as Inua Advocacy are the anchor nodes, not the government.

7. VULNERABLE POPULATIONS, DIGITAL ID RISK, AND MITIGATION

The refugee community is an exceptionally vulnerable group whose personal data demands unusual rigor — and the technology that most tempts humanitarian builders, a comprehensive digital identity for displaced people, is the one this team refuses to build. The evidence is documented, not hypothetical. UNHCR's biometric identity infrastructure (BIMS, within PRIMES) underwent a major scope shift and produced serious harm: Human Rights Watch found that biometric and biographic data collected from Rohingya refugees in Bangladesh was shared onward — ultimately reaching Myanmar, the government they had fled — for repatriation-eligibility assessment, covering some 830,000 people between 2018 and 2021, without the full data-protection impact assessment UNHCR's own policy requires and without meaningful informed consent.²¹

Consent was structurally coerced — refugees were told aid depended on registering, few could read the English consent receipts — and as late as 2025, families who refused biometric enrollment were reportedly cut off from food and services.²² The defining property of the harm is irreversibility: biometrics are “sticky” — an iris cannot be reissued once compromised — and a database built to “facilitate solutions” suffered exactly the function creep critics predicted.²³

MITIGATION IS MOSTLY REFUSAL, THEN MINIMIZATION

- **Do not build biometric identity.** A category refusal, encoded as a versioned decision-pattern in the CoLab's open schema (DDC-0001: biometric registration of displaced populations).
- **Decouple entitlement from identity** (§6.2): access gated by the thinnest credential, never “who you are” proven biometrically.
- **Data minimization & purpose limitation:** least data, single stated purpose, no secondary use or third-party transfer — by design, not policy alone.
- **Individual control & local governance:** the person holds and selectively presents any credential; refugee-led and local actors govern the data.
- **Real consent or none:** if consent cannot be free and informed — aid-conditioned, or across an unbridgeable language/power gap — treat it as absent and do not proceed.

²¹Human Rights Watch. “UN Shared Rohingya Data Without Informed Consent.” Human Rights Watch, 15 June 2021, [hrw.org/news/2021/06/15/un-shared-rohingya-data-without-informed-consent](https://www.hrw.org/news/2021/06/15/un-shared-rohingya-data-without-informed-consent).

²²“Although Shocking, the Rohingya Biometrics Scandal Is Not Surprising and Could Have Been Prevented.” ODI Insights, 2021, odi.org; “UNHCR Defends Biometric Enrollment Push for Rohingya Refugees.” The Diplomat, June 2025, thediplomat.com.

²³“Rohingya Data Protection and the UN's Betrayal.” The New Humanitarian, 21 June 2021, [thenewhumanitarian.org](https://www.thenewhumanitarian.org).

- **Mandatory pre-collection DPIA + community consultation** — the omitted step whose absence produced the Rohingya harms.

For a funder, this refusal is an asset: a responsible-AI lab that can name the one thing it will not build — and cite exactly why — is more credible than one that promises everything.

8. AGGREGATING DATASETS SO REFUGEES CAN DEMAND RESETTLEMENT SOONER

Aggregated, these instruments become an advocacy engine that attacks the core information asymmetry: individuals lack a factual, defensible basis for the rights-and-options case that today requires specialized legal and economic knowledge. LIAI shows, with disaggregation, which sub-populations are systematically failed by rights-and-pathways information — evidence, not anecdote. RICS reframes the ask from “please help” to “prolonged encampment costs X per year; this pathway costs less by year Z” — the argument donors and states respond to. The legal map converts the evidence into action, routing people to organizations that can file, refer, and litigate. Outputs are designed to be wielded by refugee-led organizations and civil-society advocates — and, where they will use them, UNHCR durable-solutions units — rather than to sit in a dashboard. Two guardrails: every advocacy artifact carries its uncertainty band honestly, and the aggregation layer never becomes a registry of identifiable individuals (§7) — it argues about populations and gaps, not named people.

9. WHAT INFRASTRUCTURE IS ACTUALLY NEEDED

LAYER	COMPONENT	REUSES	POSTURE
Measurement	LIAI index + open data standard + reference implementation + methodology	EII / composite-index method	Build — the DLI ask
Economics	RICS: encampment-vs-inclusion break-even; capacity & willingness map	ERCF engine (operational) + readiness gatekeepers	Adapt — engine in place
Value transfer	Privacy-preserving cash wallet, biometric-free	External audited network; CoLab adds threat model + measurement	Integrate, don't rebuild
Portability	Self-held skills & qualification credentials	Provenance VC architecture; ID2020 lineage	Build narrowly — skills only
Rights	Open civil-society legal-assistance map	Provenance sourcing discipline	Build — safest, high leverage
Governance	DDC decision-pattern schema; DPIA + ethics layer	DDC-0001	Build — the credibility spine

Absent by design: any biometric or identity system (§7), and any bespoke ledger (§6.2). Everything listed is either a measurement/knowledge artifact the CoLab is suited to own, or a thin integration on audited external rails — the correct risk posture for a small lab working with a highly vulnerable population.

10. HOW WE WILL MEASURE SUCCESS

A funder should not have to take this program on faith. Success is defined in advance, in three tiers — outputs (did we ship), outcomes (does it work and is it used), and impact (did it change decisions) — with the measurement instruments themselves doing double duty as the evidence base. The CoLab's standing discipline applies throughout: no meeting without a merged artifact as its deliverable, versioned releases, and public metrics.

10.1 OUTPUT INDICATORS (MONTHS 0–12): DID WE SHIP, AND IS IT RIGOROUS?

- LIAI v1.0 released open-source: index, data standard, reference implementation, and methodology documentation, each versioned and citable (DOI-minted).
- Validation completed and published: AHP/Fuzzy-AHP expert elicitation (8–12 experts, two Delphi rounds) with weights, consistency ratios ($CR < 0.10$), and divergence analysis reported — including negative results.
- Dzaleka field validation executed with FSF: information audits, structured community reporting, and a household/individual survey module; construct validity, inter-rater reliability, and test-retest statistics reported against pre-registered thresholds.
- RICS parameterized for Dzaleka on the operational ERCF engine, with every cost parameter tagged Validated/Estimated/Unvalidated and the full parameter table public.
- CSO legal-assistance map live with $\geq 90\%$ of entries verified against primary sources, and DDC-0001 published with the governance schema.

10.2 OUTCOME INDICATORS (MONTHS 6–24): DOES IT WORK, AND IS IT USED?

- Measurement performance: LIAI detects known information-access gaps at Dzaleka (criterion validity against FSF's ground truth) and its equity disaggregation (gender, age, nationality, in/out-of-camp) reproduces independently observed disparities.
- Adoption: at least one external research team or evaluation adopts LIAI as an instrument, and at least one practitioner organization (FSF or a peer) uses LIAI scores to retarget outreach — the practitioner-demand test DLI weighs.
- Use of the financial model: RICS outputs cited in at least two advocacy or policy documents by refugee-led or civil-society organizations, with uncertainty bands reproduced intact (we will audit for stripped caveats — fidelity of use is an indicator, not just volume).
- Community legitimacy: refugee community members employed as enumerators/co-designers in the validation fieldwork; a community review session at Dzaleka signs off on how findings are represented before publication.
- Open-source health: public contribution metrics (issues, forks, external commits), quarterly versioned releases, and a documented second-site adoption path.

10.3 IMPACT INDICATORS (MONTHS 12–36): DID DECISIONS CHANGE?

- A livelihoods or durable-solutions actor at Dzaleka changes a concrete allocation, outreach, or referral decision attributable to LIAI/RICS evidence (documented via decision memos or partner attestation).
- Information-access scores for the worst-served sub-populations at Dzaleka improve between baseline and endline measurement — the instrument's own longitudinal read, with the caveat that attribution is contributory, not sole-cause.

- The measurement gap itself closes: “information access” appears as a specified variable in at least one subsequent funded evaluation design that cites the LIAI standard — the definition of success for infrastructure.

10.4 WHAT A SUCCESSFUL GRANT LOOKS LIKE

At the end of the grant, a successful project looks like this: an open, validated measurement instrument exists where none did; its Dzaleka validation is published with honest statistics, including what did not work; a financial model built on an already-operational engine has produced defensible encampment-vs-inclusion figures that refugee-led advocates are actually using, uncertainty bands intact; a legal-assistance map is maintained by local partners rather than by us; no biometric or identity data was collected at any point, and the refusal is documented as a reusable governance pattern; and a second displacement context has a costed, documented adoption path. Equally important is what failure looks like, stated now: weights that fail expert consistency checks and get shipped anyway; adoption claims resting on downloads rather than documented decisions; a “map” that decays unverified; or any drift from measurement toward identity. We commit to reporting against both lists.

10.5 LEARNING & ACCOUNTABILITY MECHANICS

- Pre-registered analysis plan for the validation study; all instruments, data (de-identified, aggregate), and code published open-access at grant end, consistent with IPA data-publication guidelines.
- 8–15 project milestones and 4–7 policy-and-practice-influence milestones tracked publicly (the DLI structure), including in-country presentation of results in Malawi.
- A standing do-no-harm review: any proposed feature touching personal data is screened against §7 before development, and the screening record is public.

11. CONNECTION TO THE DLI PROPOSAL, AND HONEST ODDS

This report backs the CoLab’s Displaced Livelihoods Initiative submission (Round V, Infrastructure & Public Goods Creation): LIAI as an open measurement public good validated at Dzaleka. The fit is structural — the Infrastructure & Public Goods track invites exactly a new measure that enables future rigorous evaluations rather than a single program evaluation, and DLI’s own call names lack of information on legal status as a livelihoods barrier and invites AI-supported measurement in hard-to-reach contexts. Sylvia Maier, PhD (NYU SPS Center for Global Affairs) serves as Principal Investigator and eligibility anchor — DLI requires a PI with primary university affiliation and demonstrated experience supporting or conducting rigorous impact evaluations, so her evaluation-experience evidence must be confirmed before submission. Teresa Cantero (PhD candidate, Universidad Carlos III de Madrid — Instituto de Derechos Humanos “Gregorio Peces-Barba”; Adjunct Professor, IE University; Visiting Scholar, NYU CGA) is Primary Researcher and Co-PI; the Summer 2026 cohort carries implementation; Fraternidade Sem Fronteiras is the field partner.

Candidly: DLI’s funded portfolio skews to teams with strong impact-evaluation track records, and this is a late entry to a final round — eligible, inside the funder’s priorities, and a legitimate long shot whose competitiveness rests on measurement rigor and the team’s field depth. Items to resolve before submission: confirmation of the late-entry inquiry to displacement@poverty-action.org; the FSF letter of support; one named practitioner-demand sentence; Sylvia Maier’s consent to serve and her impact-evaluation experience evidence; and Teresa’s evaluation-experience framing (any RCT

or quasi-experimental role, stated plainly). The same architecture also serves adjacent funders — the measurement-and-governance framing travels.

12. WORK PLAN, TIMELINE, AND MILESTONES

The work plan operationalizes §§5–10 on a 24-month term with an assumed start of 1 January 2027 — inside DLI’s requirement that funded projects begin within six months of award and no later than 30 September 2027²⁴ — and follows the milestone structure §10.5 committed to: 8–15 project milestones and 4–7 policy-and-practice-influence milestones, tracked publicly. Dates below are relative to the assumed start and shift with the actual award date.

PHASED PLAN

PHASE	MONTHS	FOCUS	REPORT ANCHOR
1 — Design & partnership	M1–M4	LIAI indicator re-specification from the EII; FSF partnership agreement; IRB approval and pre-collection DPIA; AHP/Delphi expert-panel recruitment	\$6.1 · \$7 · \$3.1
2 — Build & baseline	M4–M10	LIAI reference implementation; AHP/Fuzzy-AHP elicitation (8–12 experts, two Delphi rounds); enumerator training with refugee community members; baseline information audit + household survey at Dzaleka	\$6.1 · \$10.1
3 — Validation & economics	M9–M18	Construct-validity, reliability, and equity testing against pre-registered thresholds; RICS parameterized for Dzaleka on the operational ERCF engine; CSO legal-assistance map stood up with FSF and local advocates	\$5 · \$6.4 · \$10.1
4 — Release & influence	M18–M24	Endline measurement wave; LIAI v1.0 open release (DOI-minted) with data standard, reference implementation, and methodology; in-country presentation in Malawi; second-site adoption path documented	\$10.2–10.4

PROJECT MILESTONES (DLI STRUCTURE, 12 OF 8–15)

#	MILESTONE	START	END
1	LIAI indicator framework re-specified (livelihoods & legal-rights information domains)	2027-01-01	2027-03-31
2	FSF partnership agreement, Dzaleka access protocols, IRB approval + DPIA	2027-01-01	2027-04-30
3	Instrument & survey module designed (quality / accessibility / equity)	2027-02-01	2027-04-30
4	AHP/Fuzzy-AHP expert elicitation completed; weights + consistency ratios published	2027-03-01	2027-07-31

²⁴Innovations for Poverty Action. Displaced Livelihoods Initiative: Call for Proposals and Application Instructions, Round V. IPA, 2026. Timeline and milestone requirements as restated in IPA’s DLI Eligibility Self-Assessment, 2026.

#	MILESTONE	START	END
5	Enumerator recruitment & training (refugee community members)	2027-04-01	2027-05-31
6	Instrument pilot / cognitive testing at Dzaleka	2027-05-01	2027-06-30
7	LIAI reference implementation v0.1 (open source)	2027-04-01	2027-10-31
8	Baseline information audit + household/individual survey	2027-06-01	2027-09-30
9	Construct-validity, reliability & equity analysis vs. pre-registered thresholds	2027-09-01	2028-01-31
10	RICS parameterized for Dzaleka; full parameter table public (Validated/Estimated/Unvalidated)	2027-10-01	2028-03-31
11	CSO legal-assistance map live (≥ 90% entries source-verified); endline wave	2028-02-01	2028-08-31
12	LIAI v1.0 DOI release + methodology documentation + adoption package	2028-08-01	2028-12-31

POLICY & PRACTICE INFLUENCE MILESTONES (5 OF 4–7)

#	MILESTONE	TARGET
1	Baseline findings shared with FSF for programmatic use	2027-11
2	RICS encampment-vs-inclusion figures delivered to refugee-led and civil-society advocates, uncertainty bands intact	2028-05
3	In-country presentation of results in Malawi (FSF, practitioners, durable-solutions actors)	2028-11
4	Open-access data, instruments, and code published per IPA guidelines	2028-12
5	Documented second-site adoption path released with the v1.0 package	2028-12

IMPLEMENTATION PARTNER INTEGRATION – FRATERNIDADE SEM FRONTEIRAS

Integrating FSF is a work-plan objective, not a logistical afterthought: DLI’s guidance flags researcher-only designs as unlikely to succeed on policy-influence and scalability grounds, and FSF is what makes this a researcher–implementer partnership. FSF co-designs the information domains the instrument measures, anchors community entry and enumerator recruitment, cross-checks audit findings against its operational knowledge, and is the first user of the diagnostic — the feedback loop named in the influence milestones above. This is an established relationship, not a cold one: **Yago Rocha volunteered with FSF at Dzaleka**, working directly on refugee livelihoods and financial inclusion, and **Carolina Moron visited Dzaleka while living in Malawi** building the multi-sector Malawi Voice Data Commons coalition — first-hand camp familiarity and Malawian partnership experience that de-risk community access and shorten Phase 1. Outstanding: the FSF letter of support, and formalizing the scope of FSF’s role in the Phase 1 partnership agreement.

13. BUDGET PLANNING AND ALLOCATION

The request is **US\$248,975 over two years** — US\$216,500 direct plus US\$32,475 indirect at a placeholder 15% rate (confirm NYU’s permitted rate) — just under the US\$250,000 cap DLI sets for

Infrastructure & Public Goods projects²⁵. Two disciplines govern it. First, every line maps to a work-plan phase and milestone in §12 — the budget funds the plan, nothing else. Second, DLI states that, all else equal, lower budgets have a higher probability of being funded; every line must therefore visibly earn its place, and trimming remains on the table where scope is soft.

BUDGET SUMMARY, MAPPED TO THE WORK PLAN

CATEGORY	YEAR 1	YEAR 2	TOTAL	FUNDS (WORK-PLAN ANCHOR)
Personnel (research/software lead; field research coordinator; data analyst; fringe)	\$62,000	\$46,000	\$108,000	All phases; every milestone M1–M12
Travel & communication	\$13,000	\$7,000	\$20,000	Phases 1–2 fieldwork (M5–M8); Phase 4 in-country presentation (P&P M3)
Equipment, software & programming	\$18,000	\$12,000	\$30,000	Phases 2–3 build: LIAI reference implementation (M7); RICS adaptation (M10)
Data collection — baseline	\$28,000	\$0	\$28,000	Phase 2: training, pilot, baseline audit + survey (M5–M8)
Data collection — endline	\$0	\$20,500	\$20,500	Phase 4: endline wave (M11)
Other (expert-panel honoraria, open-access release, dissemination)	\$4,000	\$6,000	\$10,000	AHP elicitation (M4); DOI release + adoption package (M12); Malawi workshop (P&P M3)
Subtotal — direct costs	\$125,000	\$91,500	\$216,500	
Indirect costs (15%, placeholder)	\$18,750	\$13,725	\$32,475	Confirm NYU negotiated rate
TOTAL	\$143,750	\$105,225	\$248,975	Cap: \$250,000

TECHNOLOGY-DEVELOPMENT ALLOCATION

Because this is an Infrastructure & Public Goods submission, the technology is the deliverable, not overhead. Direct technology-development spend is approximately **US\$67,500–74,500** — roughly 27–30% of the total: programming (US\$15,000), software, cloud and compute (US\$9,000), equipment for the build and field capture (US\$6,000), survey programming across both waves (US\$3,000), open-access release infrastructure (US\$3,500), and an allocated share of personnel time on the LIAI reference implementation and RICS adaptation (~US\$31,000–38,000; the exact percentage of the research/software lead's and analyst's time booked to engineering versus fieldwork and analysis will be stated in the submitted budget so the figure is defensible under diligence). The economics engine itself is in-kind: ERCF is already operational and validated (§3.3), which is why RICS is an adaptation line, not a build line.

Consistent with DLI's eligible-costs rule, the budget contains research and measurement costs only — no intervention-implementation or program-delivery lines. In-kind contributions not billed to DLI: the operational ERCF engine, the CoLab's provenance-credential architecture reused for the skills-credential and sourcing patterns, FSF's community access, and NYU research supervision.

²⁵Innovations for Poverty Action. DLI Eligibility Self-Assessment, 2026 (funding caps by grant type; eligible-costs rule; note that "all else equal, lower budgets have a higher probability of being funded"); IPA. DLI Budget Template, Round V, 2026 (cost-category structure).

Every cost parameter ships tagged and public, on the same Validated/Estimated/Unvalidated discipline as the models it funds (§10.1).

14. CONSOLIDATED LIMITATIONS AND FUTURE RESEARCH

- Measurement before deployment: LIAI's and RICS's weights and thresholds are provisional until expert-elicited and calibrated; nothing here is operational decision-support before that validation.
- Costs are more reliable than modeled outcomes: lead with cost and break-even (the engine's validated half); treat any modeled mortality or lives-saved figure as indicative only.
- Inclusion economics are context-dependent: the model must be able to find against inclusion where host markets are too thin, or it is advocacy, not analysis.
- Do-no-harm dominates: the documented refusal (§7) and the privacy-preserving patterns are public goods in their own right — negative results and guardrails included.
- Next steps: run the AHP/Fuzzy-AHP + calibration pipeline for LIAI at Dzaleka; parameterize RICS on Dzaleka cost inputs; stand up the CSO legal map with FSF and local advocates; publish DDC-0001 and stress-test the schema against automated benefits eligibility; resolve the DLI submission items above.